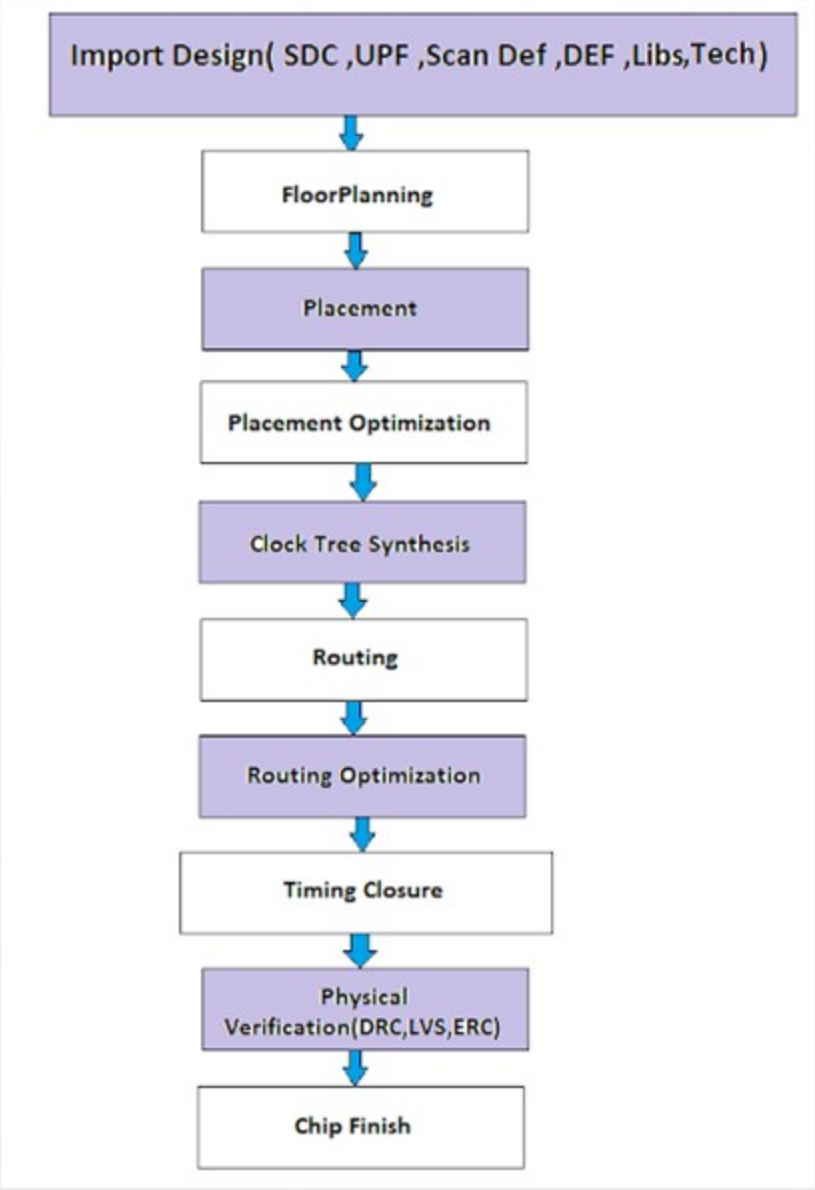
Input Files and Floor-Plan

Contents

Input Files List
Brief Explantion for each file
What is Block Level Design
Creating Core area and Die area, Parameters
Port Placement
Macros and Macro Placement
Power Planning

Physical Design Flow

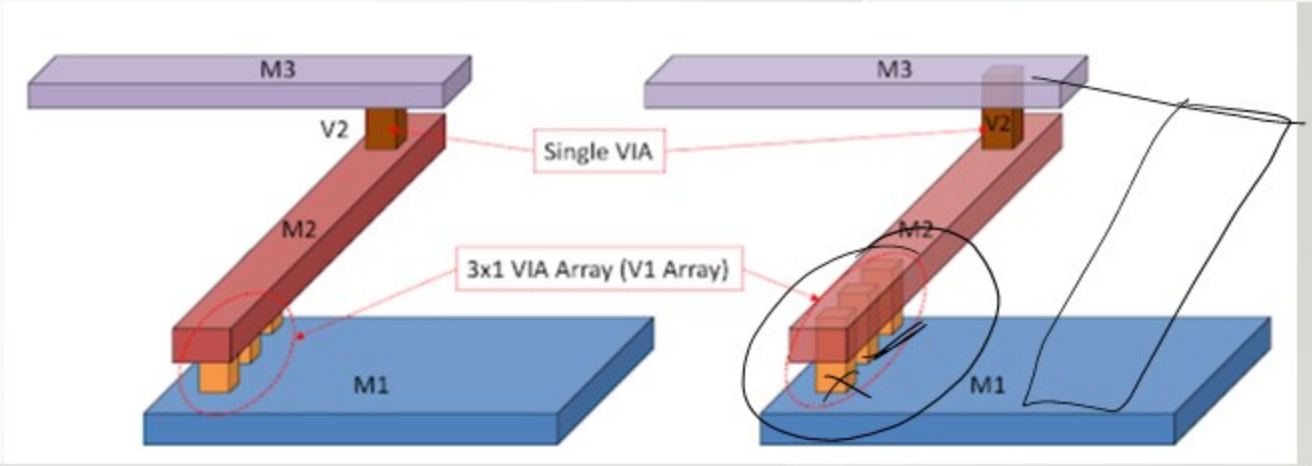


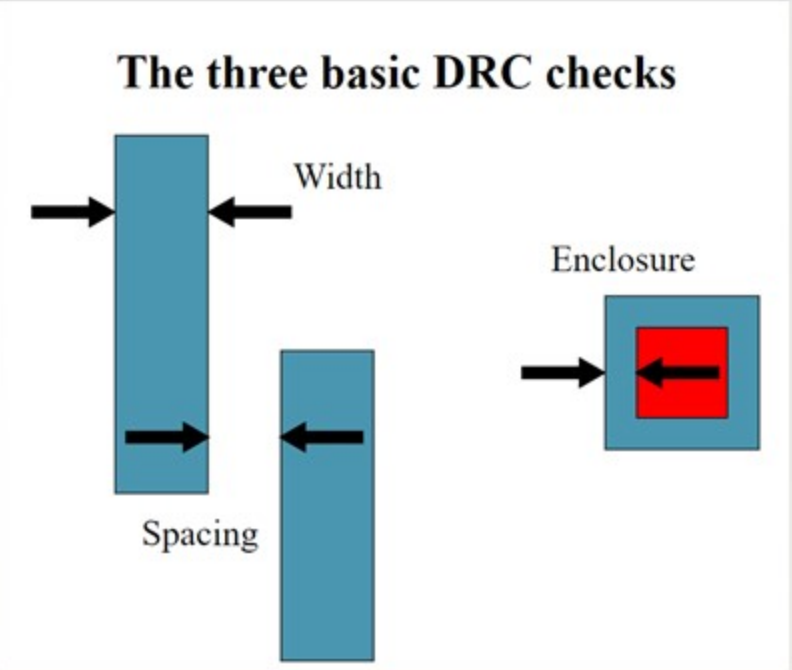
Input Files List

- Technology file [.tf]
- Tlup file [maxtlup and mintlup]
- Layout exchange format [.lef]
- Liberty timing file [.lib]
- Gate level netlist [.v]
- Synopsys design constraints [.sdc]

Technology file [.tf] :-

- It contains manufacturing grid definition and site/unit tile definition
- It contains Name, Number conventions of layers and vias
- It contains Physical, electrical characteristics of layers and vias
- In Physical characteristics Min width, Min Spacing, pitch of layers are present.
- In Electrical characteristics Max Current Density is present.





Tlup file [maxtlup and mintlup] :-

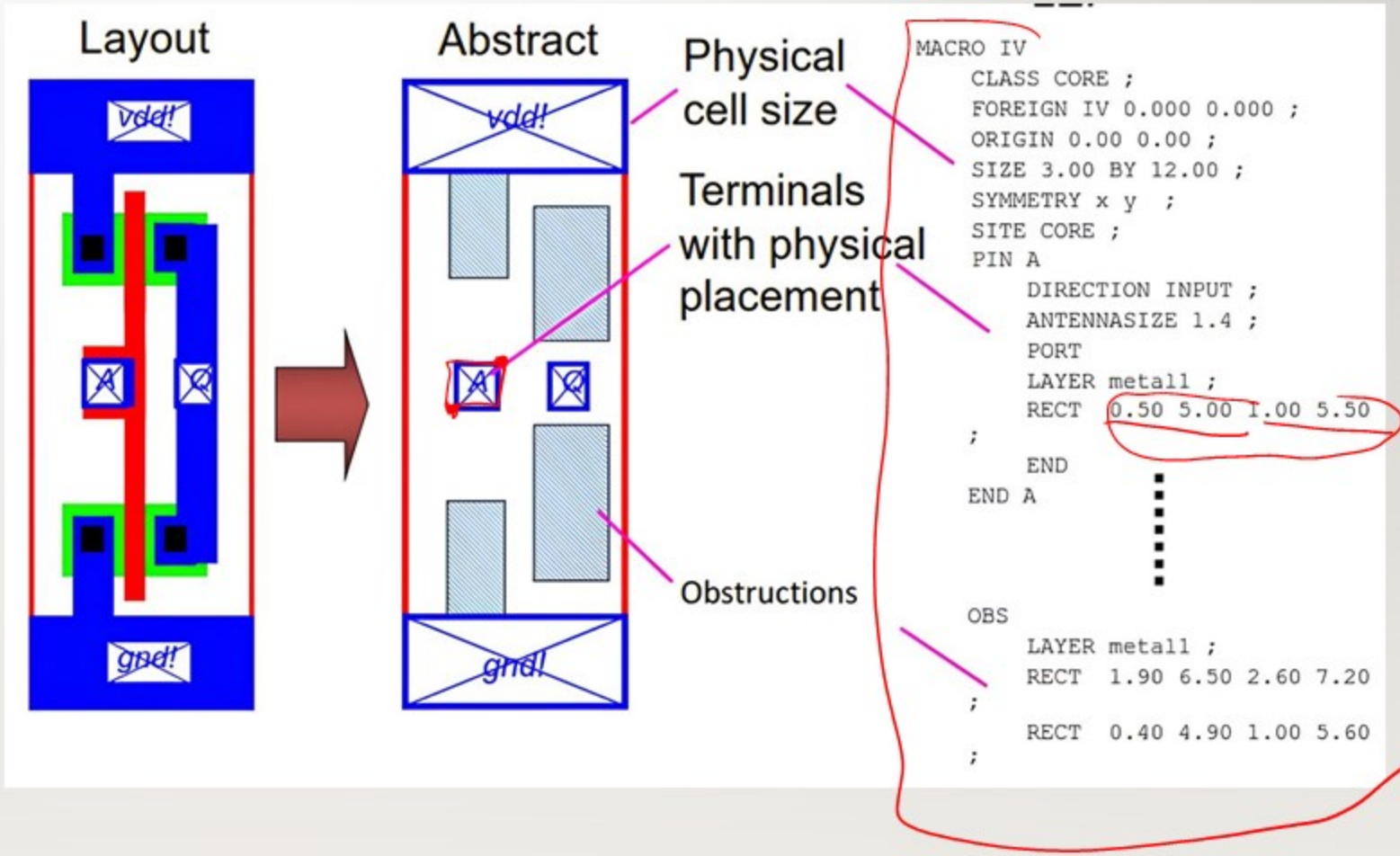
- R,C parasitic of metal layer per unit length.
- These (R,C parasitic) are used for calculating Net Delays.
- We can get TLU+ file from ITF file
- For Loading TLU+ files we have to load two files.
- Those are Max TLU+, Min TLU+

MAP file

- MAP file maps the layer and via names of TLU+ file and .tf file .

Layout exchange format [.lef]

- It contains physical information of standard cells, macros, pads.
- Contain the name of the pin, pin location (Co-ordinates) , pin layers, direction of pin (in, out, inout), uses of pin (Signal, Power, Ground) , height and width of the pin and cell.
- Size of the cell (Height and width), Symmetry of cell, Obstruction layers and its co-ordinates.



Liberty timing file [.lib]

- Timing information of Standard cells, macros.
- Functionality information of Standard cells.
- Timing DRV like max transition and max capacitance.
- In timing information look-up table is used for output transition, Cell delays, Setup, hold time.
- Cell delay is Function of input transition and output load. Cell delay is calculated based on lookup tables.
- It also has wire load model to calculate resistance and capacitance of wires.
- It also Contain Power information of Std cells.

DELAY TABLE

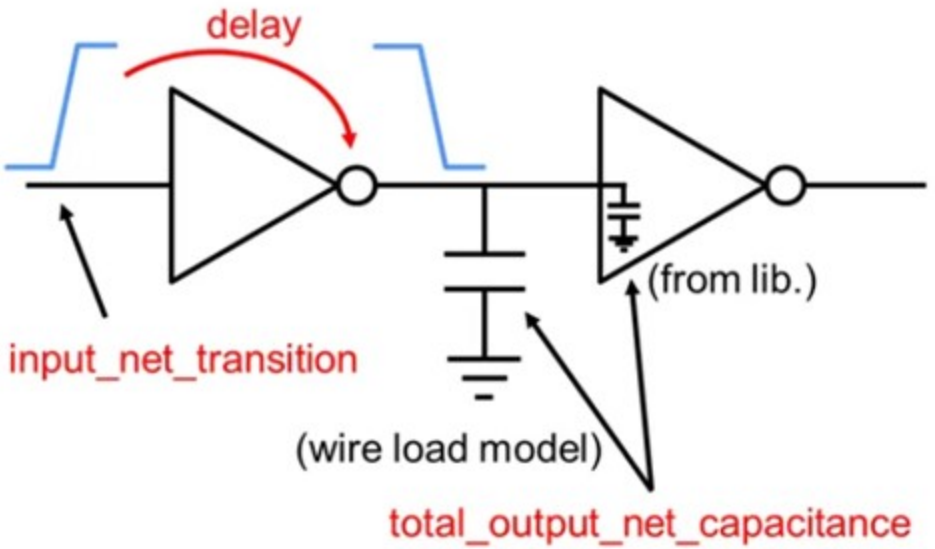
```
lu_table_template(delay_template_5x5) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  index_1 ("1000.0, 1001.0, 1002.0, 1003.0, 1004.0");
  index_2 ("1000.0, 1001.0, 1002.0, 1003.0, 1004.0");
}
```

```
cell (INVX1) {
  pin(Y) {
    timing() {
      cell_rise(delay_template_5x5) {
        values ( \
          "0.147955, 0.218038, 0.359898, 0.922746, 1.76604", \
          "0.224384, 0.292903, 0.430394, 0.991288, 1.83116", \
          "0.365378, 0.448722, 0.584275, 1.13597, 1.97017", \
          "0.462096, 0.551586, 0.70164, 1.24437, 2.08131", \
          "0.756459, 0.874246, 1.05713, 1.62898, 2.44989"); }
```

Variable_2

Variable_1

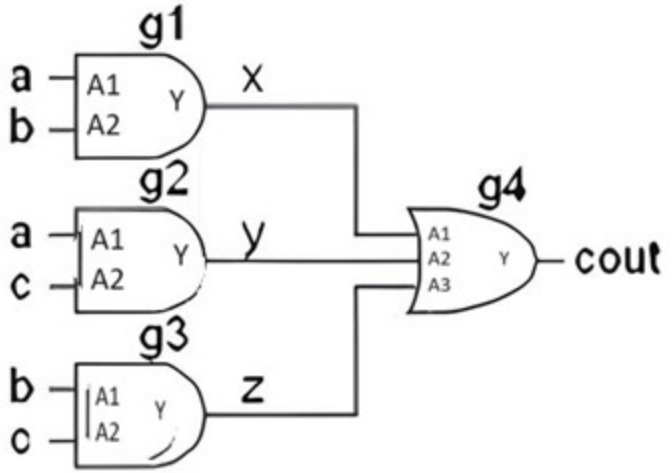
		Total_output_net_capacitance				
		1000.0	1001.0	1002.0	1003.0	1004.0
Input_net_transition	1000.0	0.147955	0.218038	0.359898	0.922746	1.76604
	1001.0	0.224384	0.292903	0.430394	0.991288	1.83116
	1002.0	0.365378	0.448722	0.584275	1.13597	1.97017
	1003.0	0.462096	0.551586	0.70164	1.24437	2.08131
	1004.0	0.756459	0.874246	1.05713	1.62898	2.44989



Cell Delay = func(input_net_transition, total_output_net_capacitance)

Gate level netlist [.v]

- It contains logical connectivity of all cells (Std cells, Macros).
- It contains list of nets Connecting std cells and Macros.
- Each cell has its own instance/cell name and library/ref name.




```
module carry (input a, b, c, output cout)

wire x , y , z;

AND2 g1 {.A1(a) , .A2(b) , .Y (x));

AND2 g2 {.A1(a) , .A2(c) , .Y (y));

AND2 g3 {.A1(b) , .A2(c) , .Y (z));

OR2 g4 {.A1(x) , .A2(y) , .A3(z) , .Y (cout));
```

Synopsys design constraints [.sdc]

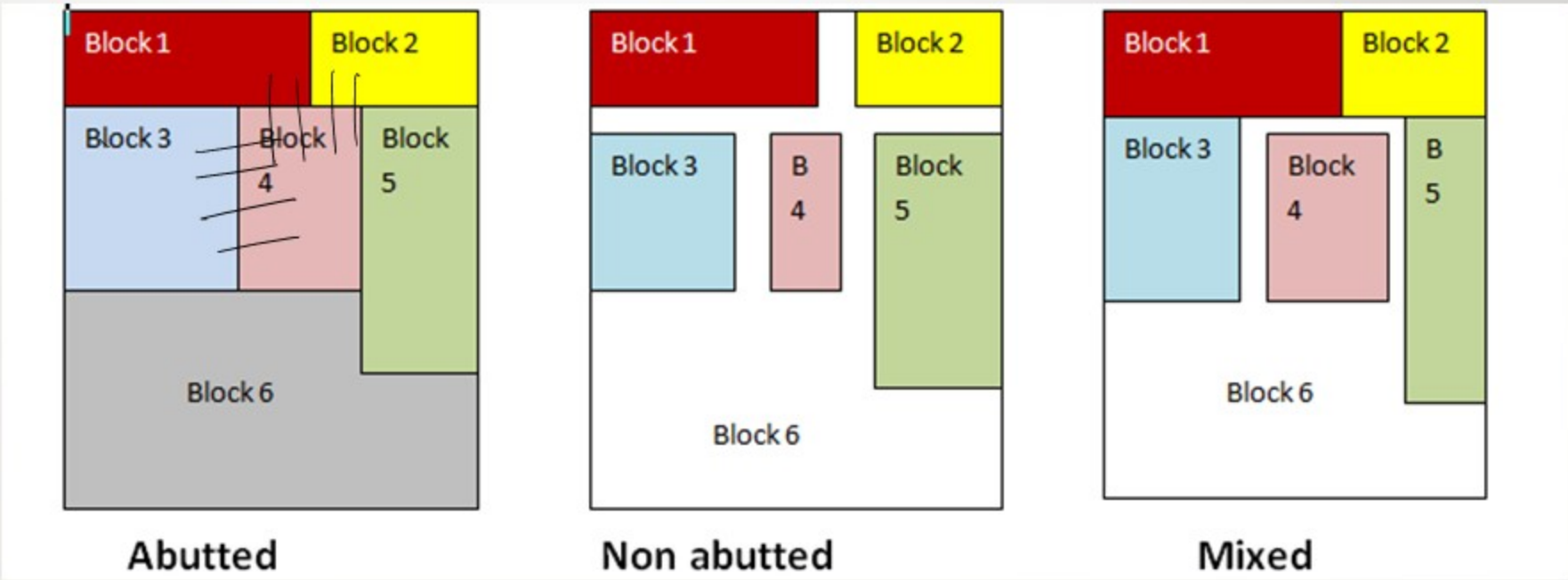
- These Constraints are mainly used to meet timing requirements of design .

- CLOCK DEFINITIONS: To create different types of clocks
- Clock uncertainty
- Setting Input Delay, Setting Output Delay, Driving Cell, Setting load on output ports
- Timing Exceptions :
  Multi cycle path, False path, Half cycle path, Disable timing arcs, Case Analysis

Fig. sdc file example

What is Block Level Design

- Large chips with millions of cells is divided in to small sub systems called as blocks
- Dividing a full chip in to blocks is called as partitioning



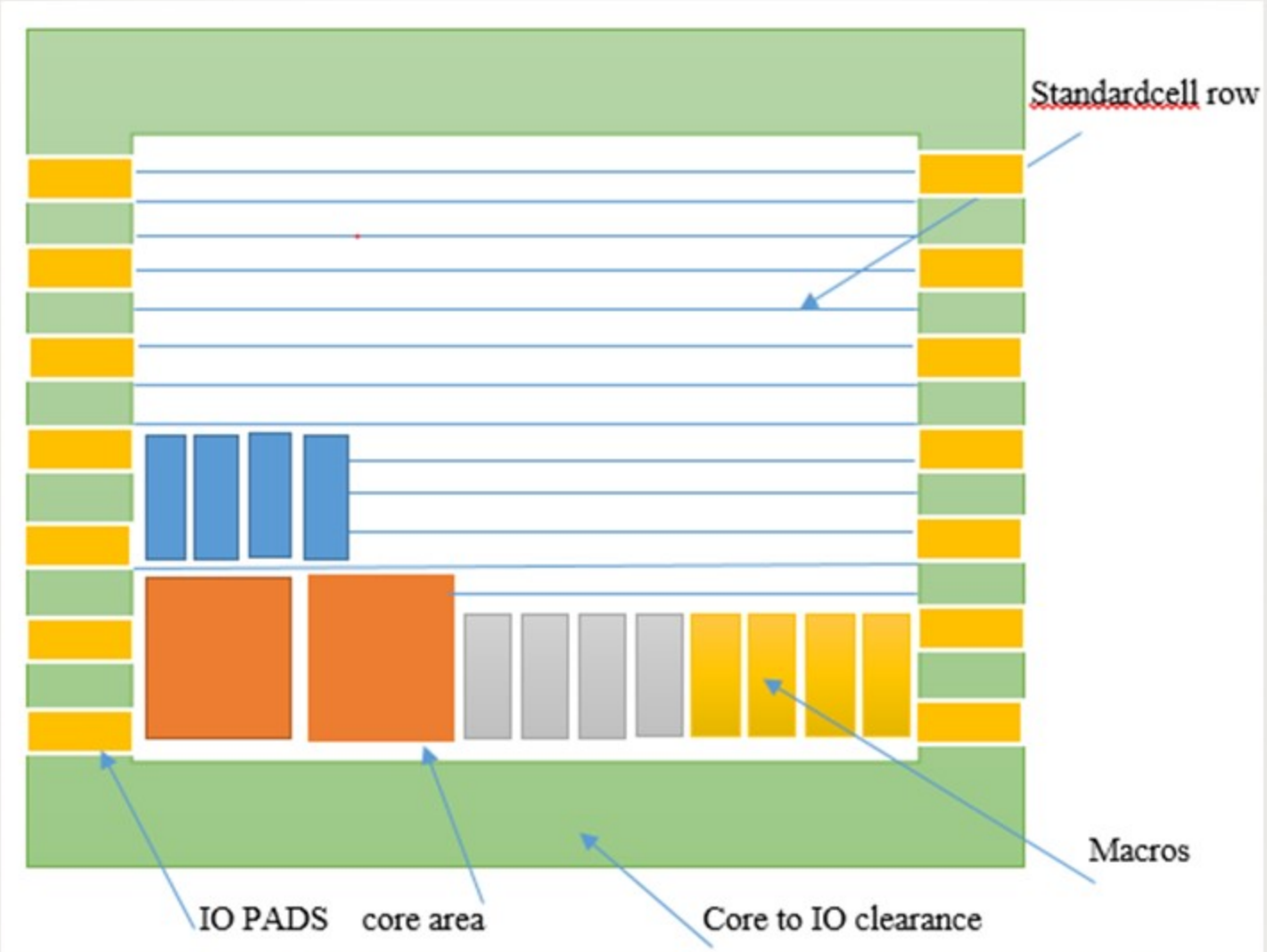
Floorplan

Floor-plan design to it is process of creating an area for macros and standard cells to be placed in efficient way

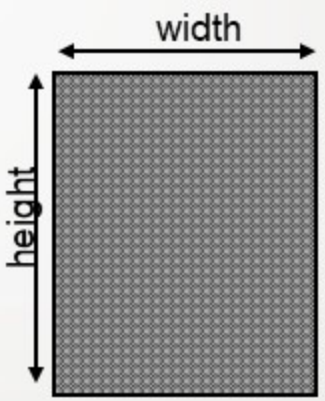
Floor Planning Steps :

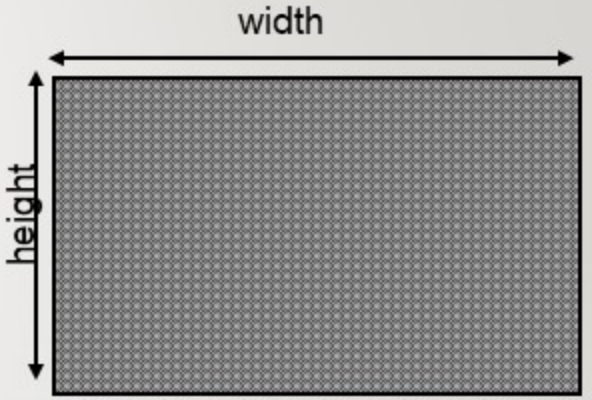
- Decide core width and height for die size estimation.
- Placement of IO pads/Ports.
- Creating Voltage area.
- Placement of macros.
- Adding physical only cells.
- Power planning (pre routing)

Creating Core area and Die area, Parameters



Core area depends upon





Aspect ratio : -

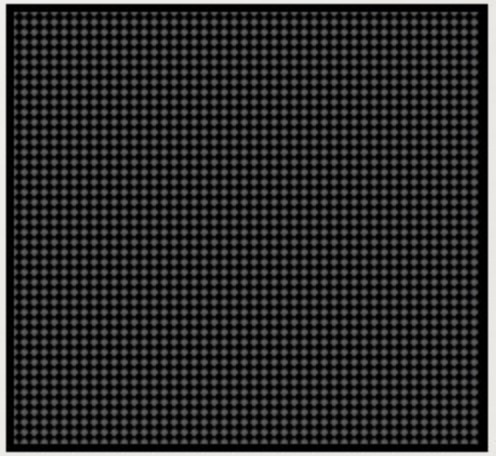
Aspect ratio will decide the size and shape of the chip . It is ratio of height and width of core area.

Aspect ratio = Width / Height

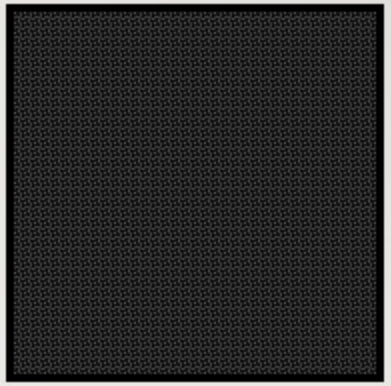
Core utilization : -

Utilization will define the area occupied by the standard cells, macros, and other cells. If core utilization is 0.8 (80%) that means 80% of the core area is used for placing the standard cells and macros and the remaining 20% is free.

$$\text{core utilization} = \frac{(\text{macros area} + \text{std cell area})}{\text{Total core area}}$$



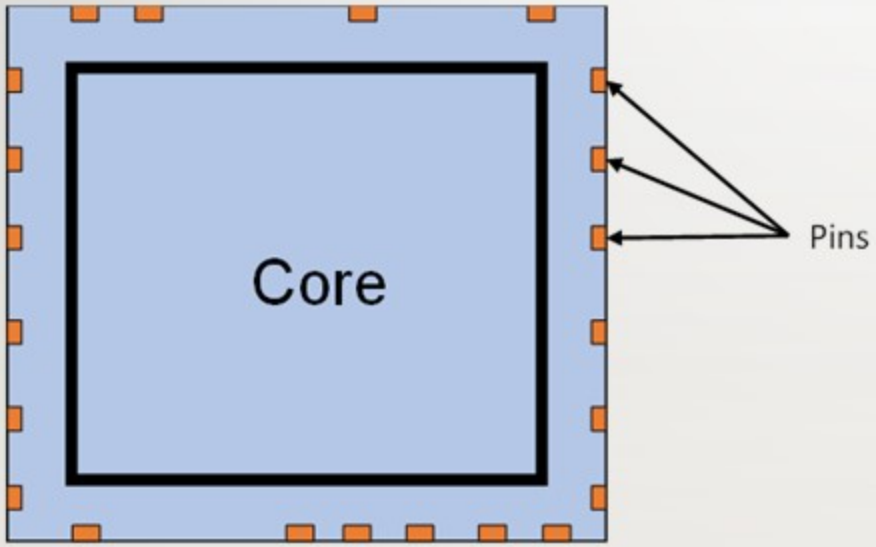
core_utilization 0.6

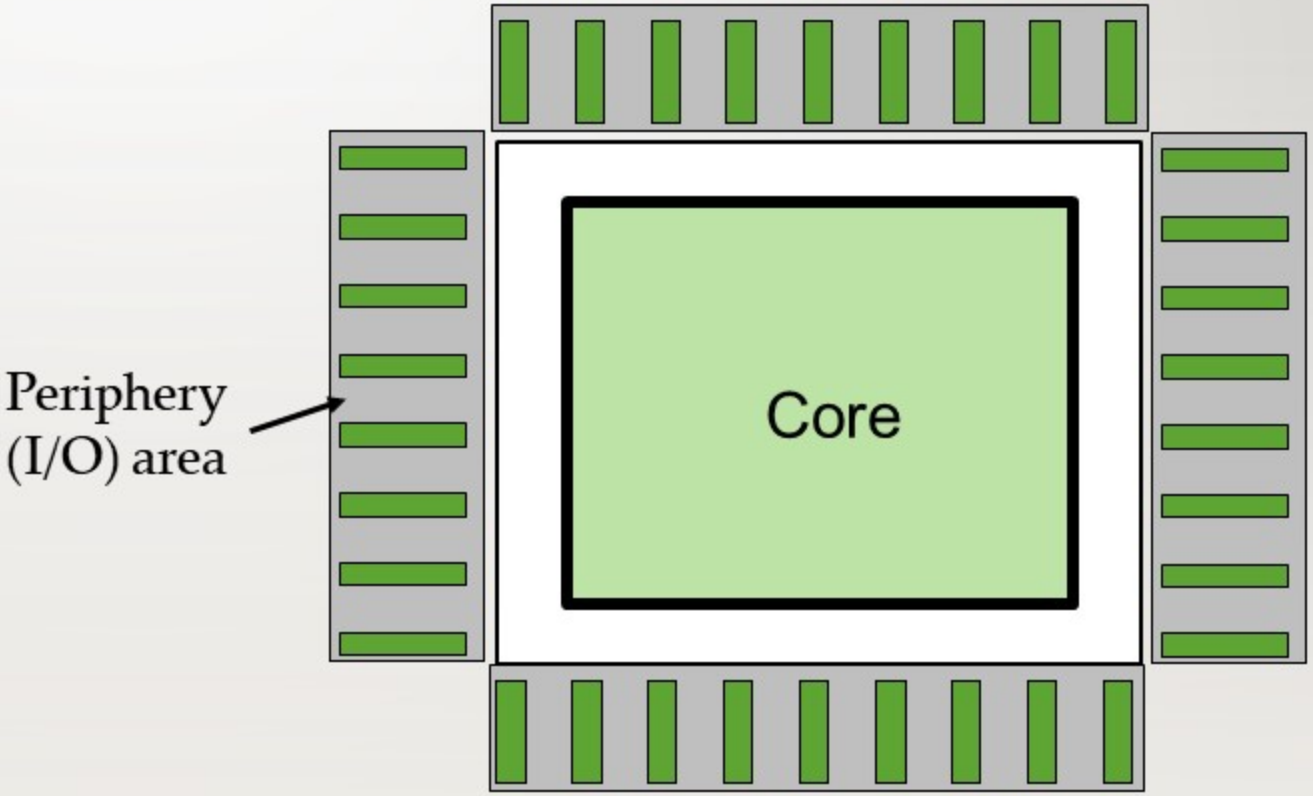


core_utilization 0.85

Port Placement

- Pins can be spread automatically by IC Compiler around the boundary.
- The exact side and/or location, layers, size for pins can be set by a designer.





Macros and Macro Placement

Macros are specially designed cells. Macros may be memories, analog blocks. Proper placement of macros has a great impact on the quality and performance of the ASIC design. Macro placement can be manual or automatic. Generally manual macro placement is Preferred.

Types of macros :-

- Hard macros - The circuit is fixed. We can't see the functionality information about macros. Only we know the timing information.
- Soft macros - The circuit is not fixed and we can see the functionality and type of gates present in it, also we know the timing information.

Basic constraints :-

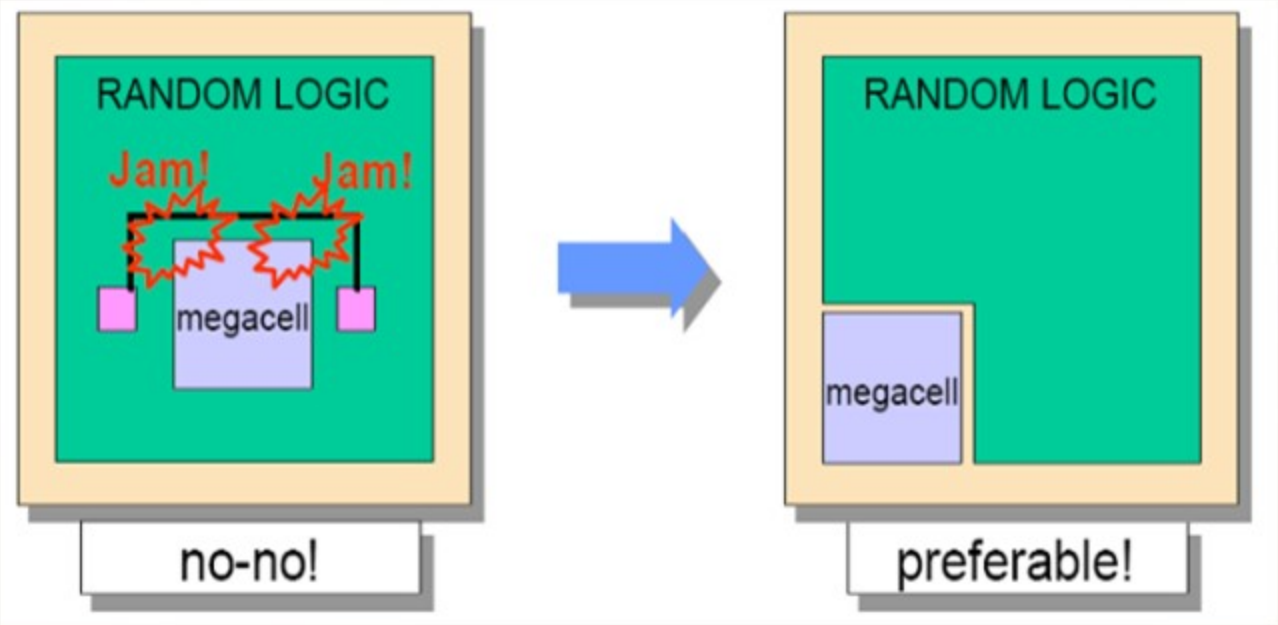
- Alignment of macros by edges.
- Grouping macros in a way that for standard cells rectangular area should be left as much as possible
- Alignment of macros with core boundary.

Goals of Macros Placement :-

- Better Rout ability
- Timing
- Reduced Wire length
- Area for standard cells

Macro Placement Guidelines

1. Macros should be placed at the periphery of the block.
   Avoid placement of Macros or stacks near the ports in order to avoid congestion in later stages of physical design.



2. Placement of macros are the based on the fly-lines (Fly-lines shows the connectivity b/w macro to macro and macro to ports /pins) so we can minimize the interconnect length between IO ports , macros and other cells.

macros to macros fly lines:

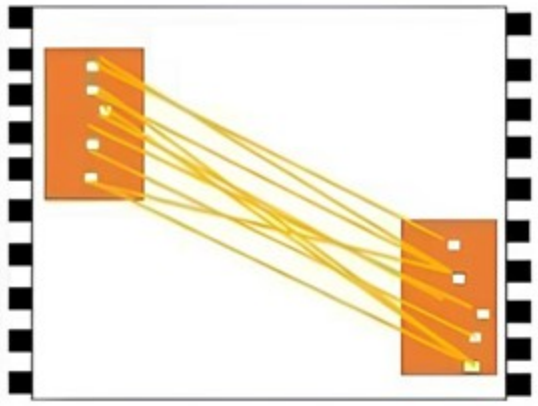


Fig a: macro to macro fly lines

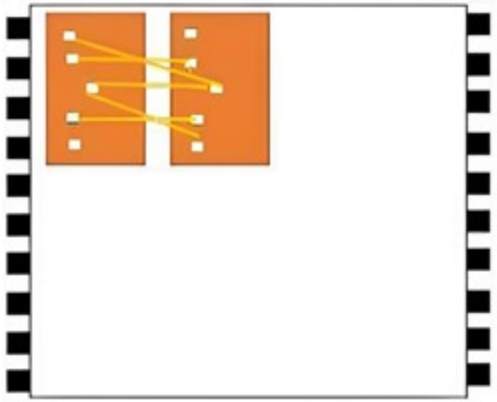


fig b: macros placed near to each other

macros to IO pin:

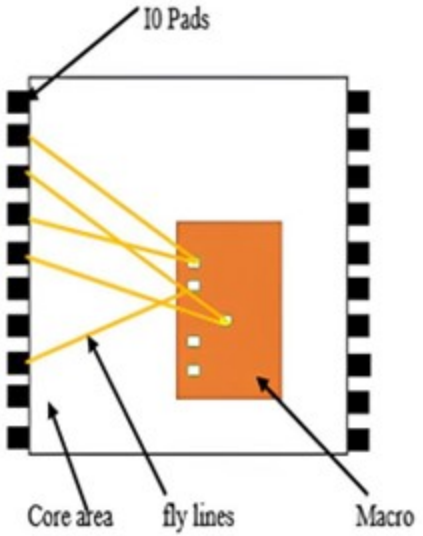


Fig a: macro to IO ports fly lines

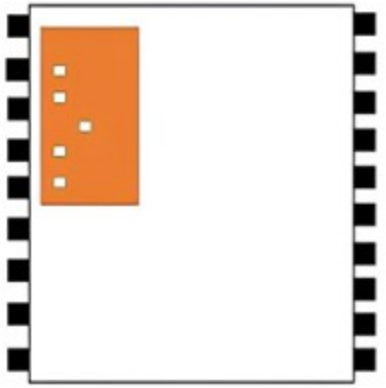
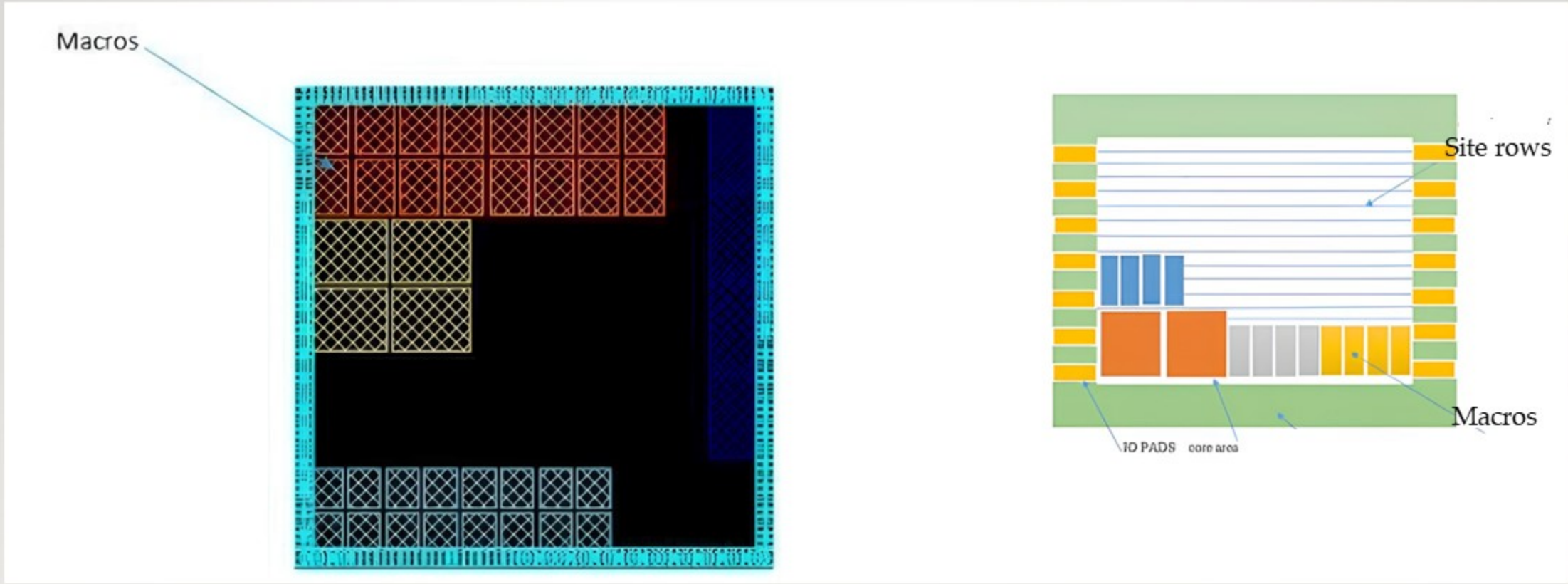
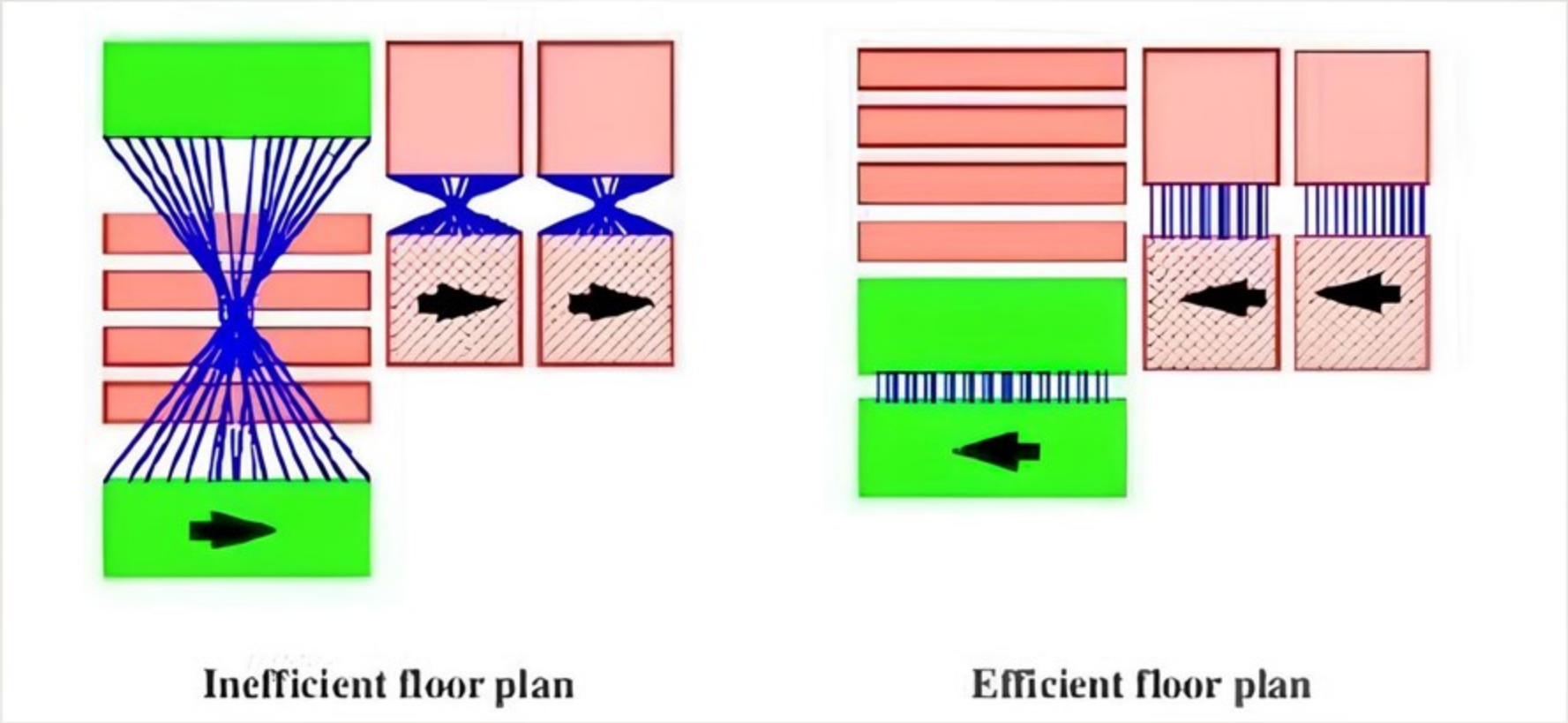


Fig b: macro placed at the core boundary

3. Place the macros of same hierarchy together



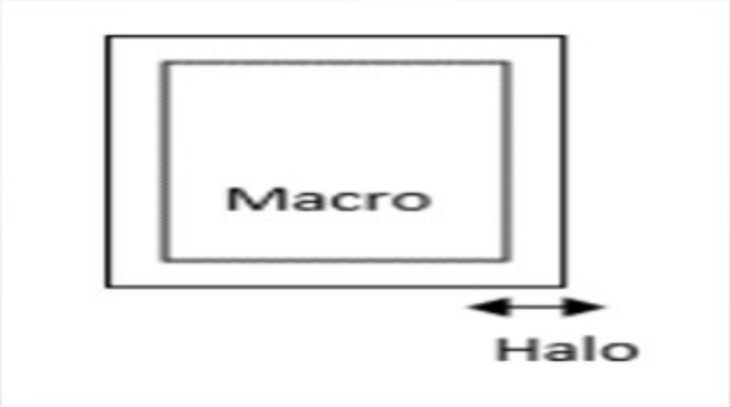
4. Macros should be placed by taking care of channel optimization to avoid crisscross placement of Macros.
5. Placement should be done in proper multiple of tracks so that macros can be routed properly.
6. Placement of macros should happen in a way so that Macro pins face towards core to avoid detour.



Inefficient floor plan

Efficient floor plan

7. After macro placement, we need to place halos/keepout margins around the Macros to avoid congestion around the macros.



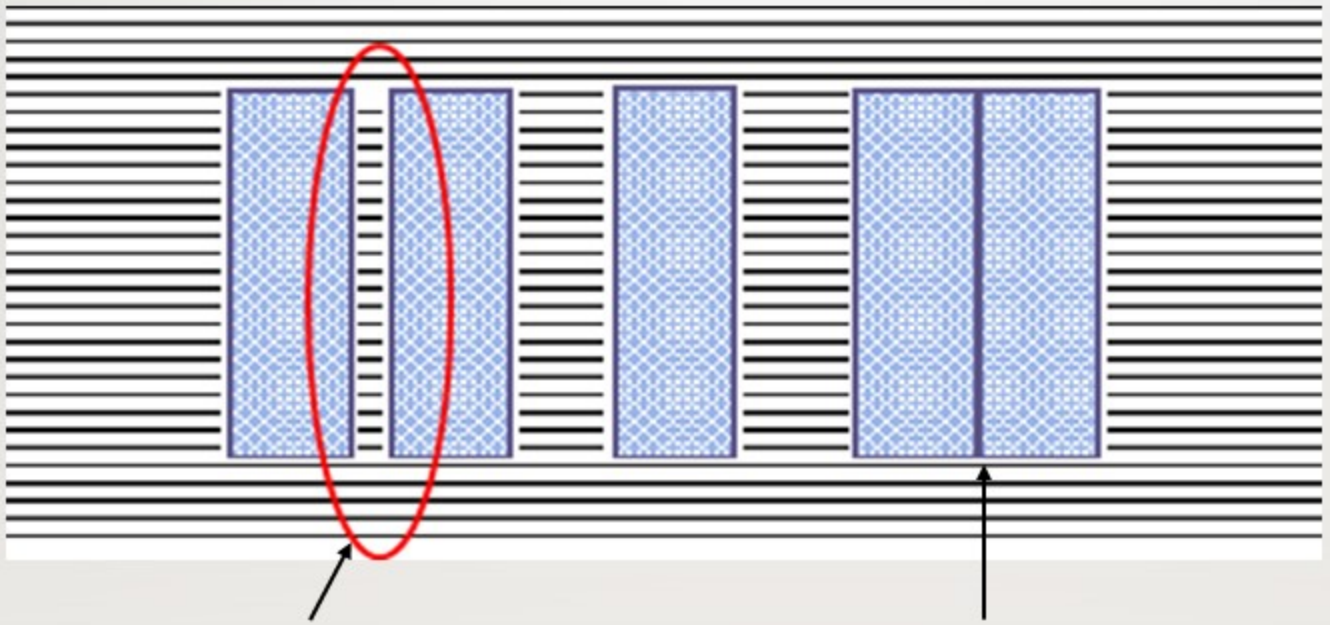
8. Placement blockages are applied between macro channel, so that no standard cell should sit in the channel, in order to avoid routing congestion.
9. Avoid stacking of macros

10. Try to avoid notches while placement of macros.

11. There should be minimum gap between the Macros.

*Distance between macros = (No. of Pins * pitch *2) / Available metal Layers*

We should allow channels for routing pins and for buffer insertion which helps during timing optimization while placing possible required buffer.



Physical cells

- In Physical design : to avoid certain violations like latch up, IR drop etc. and to meet chip manufacturing criteria we use Physical cells [also we call it as Physical only cells].

- This cells does not have any logical functionality.

Types of Physical cells :

- Endcap / boundary cells
- Well Tap cells
- De-cap cells
- Tie cells [tie high and tie low cells]
- Spare cells
- Filler cells

Physical cells Placement

VLSIGURU

Endcap / boundary cells :

- To show end of site rows and to avoid damage to the cells during manufacturing of chip.
- Placed during floorplan stage.

Well Tap cells :

- To avoid latch up problem we are placing at certain distance.
- Placed during floorplan stage.

De-cap cells :

- To avoid IR drop in design and its placed during power plan stage.

Tie cells [tie high and tie low cells] :

- To protect gate terminal of cells from input fluctuation.
- Helps to connect gate of cells to logic 1 or 0.
- Placed during placement stage.

Spare cells :

- To perform ECO in design.
- Placed during placement stage and in few cases placed during routing stage.

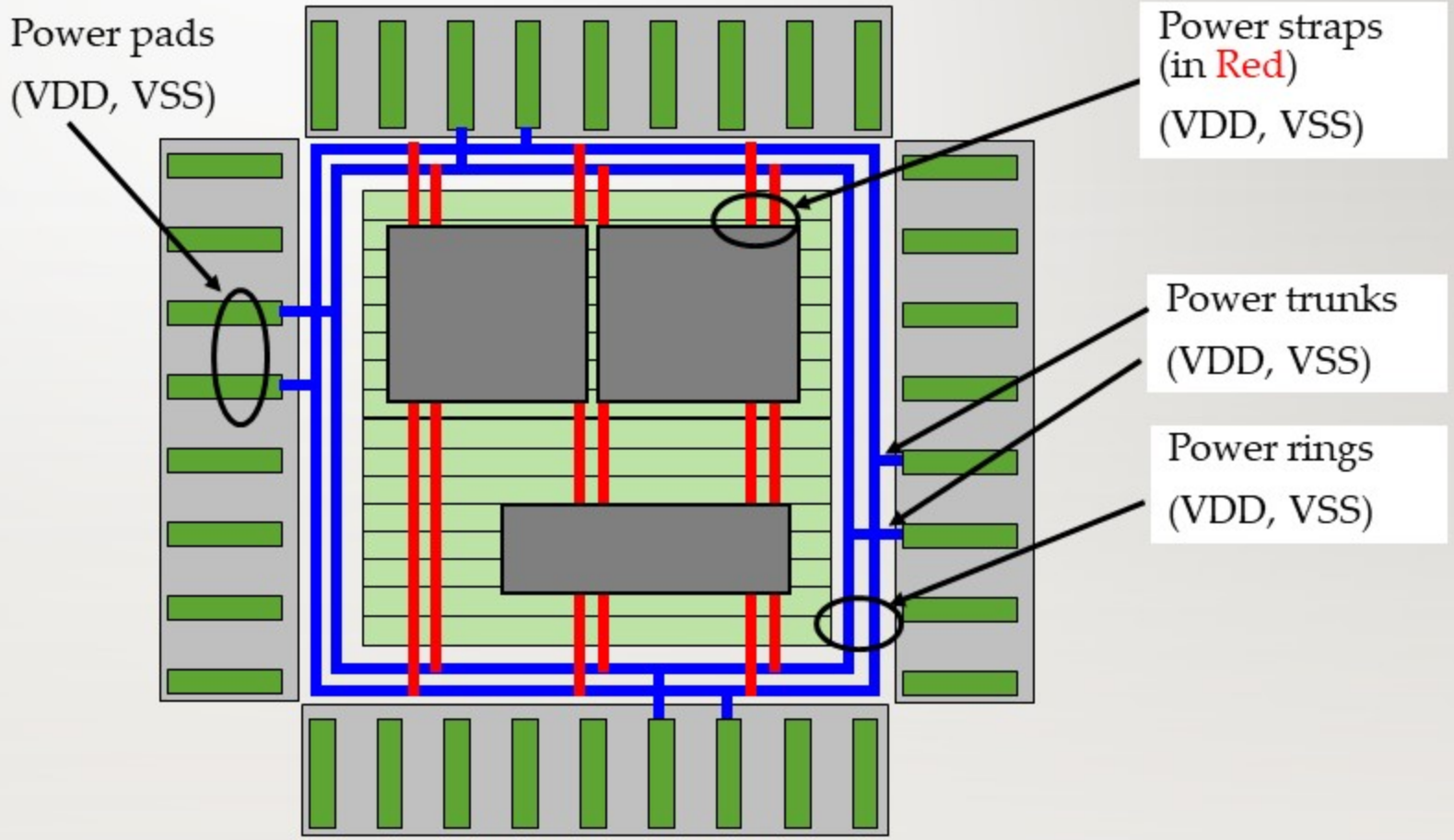
Filler cells :

- To fill empty gaps in chip placed after completing all signoff stage and before moving to chip fabrication.

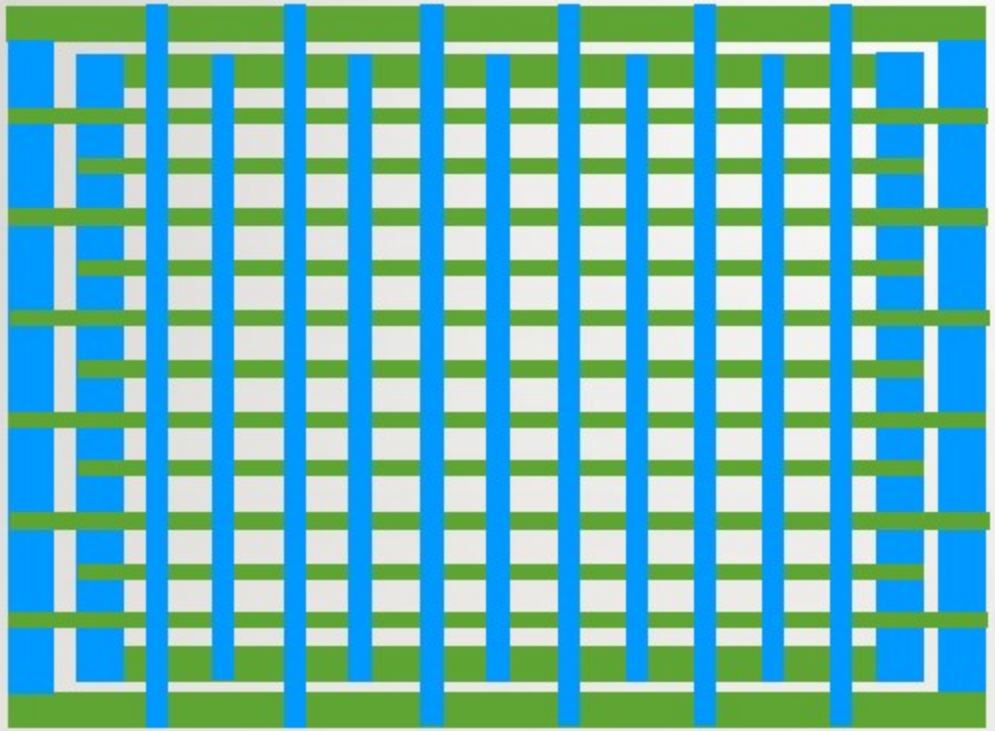
Power Planning

Power structure includes

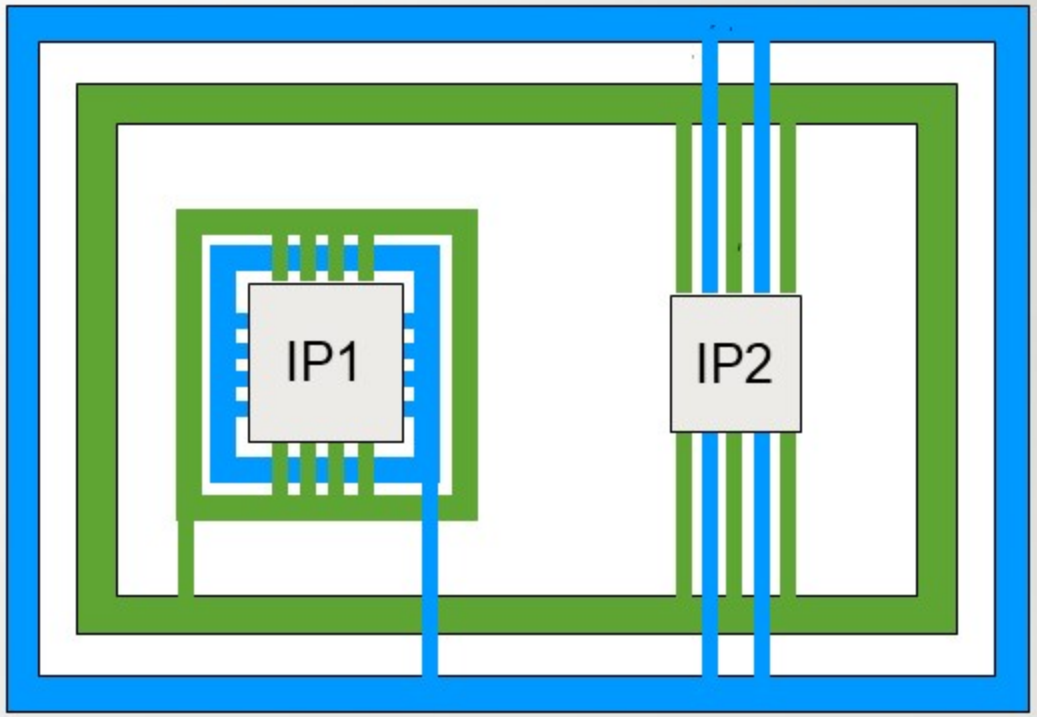
- Rings [block and macros]
- Trunks [to connect power pads to block rings]
- Straps / strips / grids
  upper straps – to connect between rings to lower straps
  lower straps – to connect between upper straps to rails
- Rails
  to connect between lower straps to std cells power pins



Top level Power Planning

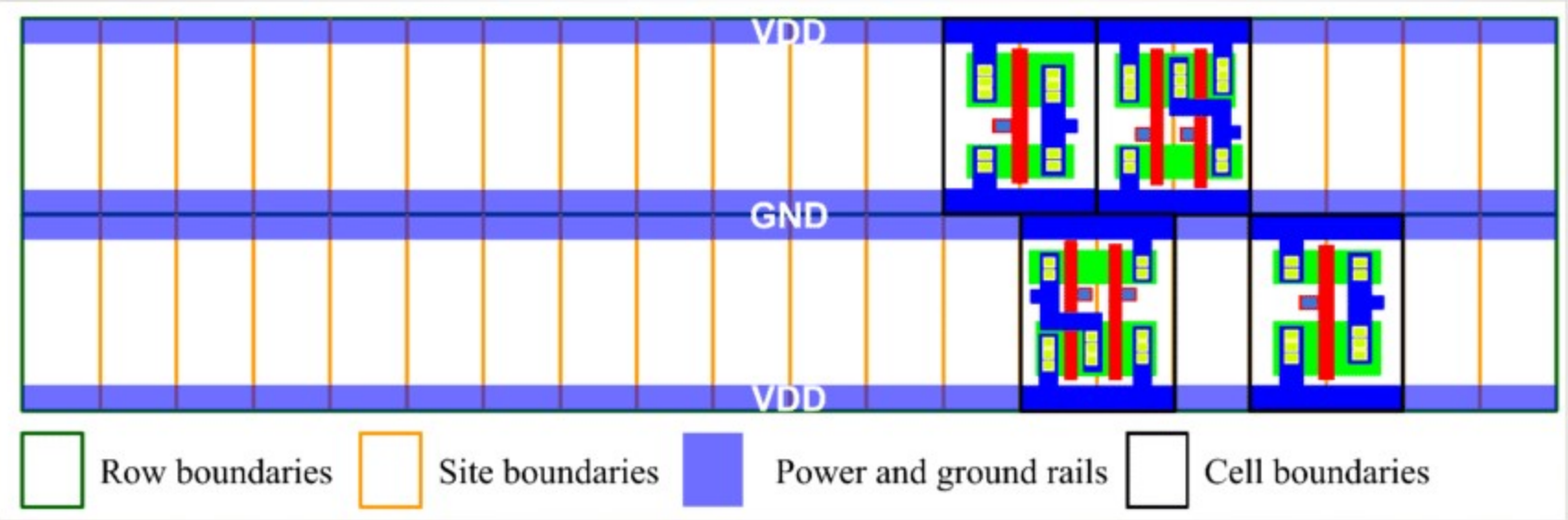


Block level Power Planning



Designer can specify

- Number of straps: Min, max
- Width of straps: Min, Max
- Width of ring
- Layers



- Rails connecting to standard cells

Steps in Power Planning

- Determine the number of power pads
- Determine which metal layers will be used for power routing
- Define the width of the top-level power bus
- Determine the structure for block and macro-level power routing
- Power Network Analysis (PNA)

Block level Power Planning

- Straps are Formed by using M7 and M8 layers
- Lower level straps are created in M2 layer
- M2 layer get connected to M7
- Rails connect to lower level straps .
- Macro rings are created around hard macro using M5 and M6 Layer which connects to straps M7 and M8
- Macro pin connects to macro rings

Analysis after Power Planning

- **Check PG DRC : Verify all DRC rules is met in power grid**
- **Check PG connectivity :**
  - **Verify floating rails**
  - **Verify floating macro pins**
  - **Verify floating macro rings**
  - **Verify floating macro pins**
- Check pg missing Vias